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§2.5 Irreducible $(\mathfrak{g}, K)$ -Modules for $GL(2, \mathbb{R})$

of Automorphic Forms and Representations by Daniel Bump

Original Goal: Compute the unitary dual of $GL(2, \mathbb{R})$, i.e., classify irreducible unitary representations of $GL(2, \mathbb{R})$ up to isomorphism.

What we will achieve today: Compute the admissible dual of $GL(2, \mathbb{R})$, i.e., classify irreducible admissible representations of $GL(2, \mathbb{R})$ up to infinitesimal equivalence.

Notation: $G = GL(2, \mathbb{R})^+$, $K = SO(2)$, $\mathfrak{g}_0 = \mathfrak{gl}(2, \mathbb{R})$, $\mathfrak{g} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C} = \mathfrak{gl}(2, \mathbb{C})$.

$$\hat{H} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \hat{R} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \hat{L} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, H = -i \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, R = \frac{1}{2} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, L = \frac{1}{2} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}, W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

elements of $\mathfrak{g} = \mathfrak{gl}(2, \mathbb{C})$.

$$\Delta = -\frac{1}{4}(\hat{H}\hat{H} + 2\hat{R}\hat{L} + 2\hat{L}\hat{R}) \in Z(\mathcal{U}(\mathfrak{g}_0)) \quad \cdot \quad \sigma_k(K_0) = e^{ik\theta}, \quad K_0 = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

$$= -\frac{1}{4}(HH + 2RL + 2LR) \quad (\text{Casimir})$$

irred unitary rep of $K = SO(2)$ are param by integers $k \in \mathbb{Z}$. character σ_k .

Proposition 2.5.1 (Dixmier's Lemma): Let V be an irreducible admissible $(\mathfrak{g}_0, K)$ -module for $GL(2, \mathbb{R})^+$. Let $D \in Z(\mathcal{U}(\mathfrak{g}))$. Then D acts by a scalar on V .

→ upshot: Both $Z = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\Delta \in Z(\mathcal{U}(\mathfrak{g}))$ act as scalars on V .

Notation: Write $\lambda \in \mathbb{C}$ and $\mu \in \mathbb{C}$ for the respective eigenvalues of Δ and Z on V .

→ we will classify irred. adm. $(\mathfrak{g}_0, K)$ modules using these invariants.

Proposition 2.5.2: Let V be an irreducible admissible $(\mathfrak{g}_0, K)$ -module for $GL(2, \mathbb{R})^+$.

(i) $V(k) = \{x \in V \mid Hx = kx\}$ ($V(k)$ is the σ_k -isotypic component, so V admissible $(\mathfrak{g}_0, K)$ -module $\Rightarrow V = \bigoplus_{k \in \mathbb{Z}} V(k)$ with $\dim_{\mathbb{C}} V(k) < \infty$ for all k)

(ii) If $x \in V(k)$, then $Rx \in V(k+2)$, $Lx \in V(k-2)$.

(iii) If $x \in V(k) \setminus \{0\}$, then $\mathbb{C}x = V(k)$, $\mathbb{C}R^n x = V(k+2n)$, $\mathbb{C}L^n x = V(k-2n)$,

$$V = \mathbb{C}x \oplus \bigoplus_{n>0} \mathbb{C}R^n x \oplus \bigoplus_{n>0} \mathbb{C}L^n x$$

(iv) $\dim_{\mathbb{C}} V(k) \leq 1$. The K -types of V are either all odd or all even.

(v) $LRx = (-\lambda - \frac{k}{2}(1 + \frac{k}{2}))x$, $RLx = (-\lambda + \frac{k}{2}(1 - \frac{k}{2}))x$.

(vi) If $x \in V(k) \setminus \{0\}$ and $Rx = 0$, then $\lambda = -\frac{k}{2}(1 + \frac{k}{2})$.

If $x \in V(k) \setminus \{0\}$ and $Lx = 0$, then $\lambda = \frac{k}{2}(1 - \frac{k}{2})$.

(vii) Suppose $\lambda = \frac{k}{2}(1 - \frac{k}{2})$ and $x \in V(1)$. If $Rx = 0$, then either $l = -K$ or $l = k$.
 If $Lx = 0$, then either $l = K$ or $l = 2-k$.

Proof: (i). Highest weight theory.

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(ii) Compute action of root vectors on weight vectors.

(iii) Use irreducibility of V and show RHS is invariant under $\mathfrak{g}$ & K .

$$\text{Use } -4\Delta = H^2 + 2RL + 2LR$$

$$\hookrightarrow \text{Idem} = H^2 + 2H + 4LR$$

$$= H^2 + 2H + 4RL$$

$$a) [R, L] = H$$

$$(H = RL - LR \Rightarrow H + LR = RL)$$

(iv) $\Leftarrow$ (iii)

(v) $\Leftarrow$ (vi) $\Leftarrow$ (vii) $\Leftarrow$ (vi) $\square$

Theorem 2.5.1: Let $\lambda, \mu \in \mathbb{C}$. Assume $\lambda \neq \frac{k}{2}(1 - \frac{k}{2})$ for some even (resp. odd).

Then there exists at most one isomorphism class of even (resp. odd).

$(\mathfrak{g}_0, K)$ -modules V on which Δ and Z have eigenvalues λ and μ resp.

For such a V , the set of K -types of V consists of all even (resp. odd) k .

Proof: To see $V(k) \neq 0$ for all even (resp. odd) k use Prop 2.5.2(iii) & (vi).

Proof of uniqueness: Let V' be another such $(\mathfrak{g}_0, K)$ -module.

WTS: There exists an isomorphism $V \rightarrow V'$.

Let $\varepsilon = 0$ or 1 (if even or odd resp), pick $x \in V(\varepsilon) \setminus \{0\}$.

Define linear isomorphism $\phi: V \rightarrow V'$ by $x \mapsto x'$, $x' \in V'(\varepsilon) \setminus \{0\}$.

Check $\phi \circ X = X \circ \phi$ for all $X \in \mathfrak{g}$. $\square$

Theorem 2.5.2: Let $k \geq 1$, integer, $\lambda = \frac{k}{2}(1 - \frac{k}{2})$. Let V be an irred.

admissible $(\mathfrak{g}_0, K)$ -module with same parity as k . Let Σ be the set

of K -types of V . The Σ is one of the following sets:

$$\Sigma^+(k) = \{l \in \mathbb{Z} \mid l \equiv k \pmod{2}, l \geq k\}$$

$$\Sigma^-(k) = \{l \in \mathbb{Z} \mid l \equiv k \pmod{2}, l \leq -k\}$$

$$\Sigma^0(k) = \{l \in \mathbb{Z} \mid l \equiv k \pmod{2}, -k \leq l \leq k\}.$$

There exists at most one isomorphism class of $(\mathfrak{g}_0, K)$ -module with each

given Σ . (The case of $\Sigma = \Sigma^0(k)$ cannot occur if $k = 1$)

Proof: Case by case analysis using Prop 2.5.2. Uniqueness similar argument as before.

Theorem 2.5.4 (Classification of $(\mathfrak{g}_0, k)$ -modules for $GL(r, \mathbb{R})^+$). The following is a complete

list of the irreducible admissible $(\mathfrak{g}_0, K)$ -modules when $\mathfrak{g}_0 = \mathfrak{gl}(r, \mathbb{R})$, $K = SO(r)$.

Every irreducible admissible $(\mathfrak{g}_0, K)$ -module may be realized as the space of K -finite

vectors in an admissible representation of $G = GL(r, \mathbb{R})^+$ on a Hilbert space.

Let $\lambda, \mu \in \mathbb{C}$ and let $\varepsilon = 0$ or 1 .

(i) If $\lambda \neq \frac{k}{2}(1 - \frac{k}{2})$, $k \equiv \varepsilon(2)$, $\exists!$ irred. adm. $(\mathfrak{g}_0, K)$ -mod. $\pi_{\lambda, \mu, \varepsilon}$ on which Δ

and Z act as λ and μ . The set of K -types is $\mathbb{Z} + \varepsilon$.

(ii) If $\lambda = \frac{k}{2}(1 - \frac{k}{2})$, $k \equiv \varepsilon(2)$, $k \geq 1$, $\exists 3$ irred. adm. $(\mathfrak{g}_0, K)$ -mod. $\pi_{\lambda, \mu, \varepsilon}$ on which Δ

and Z act as λ and μ , except if $k=1$, then $\exists 2$. K -types are $\Sigma^{\pm}(k)$ and (if $k \geq 2$) $\Sigma^0(k)$.

(3) Existence.

Notation: $\varepsilon = 0$ or 1 . $s_1, s_2 \in \mathbb{C}$ parameters.

→ We will construct an admissible representation $\pi(s_1, s_2, \varepsilon)$ of $G = GL(2, \mathbb{R})^+$ on a Hilbert space.

- $\lambda = s(1-s)$ ← eigenvalue of Δ λ "spectral parameter"
- $s = \frac{1}{2}(s_1 - s_2 + 1)$
- $\mu = s_1 + s_2$ ← eigenvalue of Z (s_1, s_2) "Langlands parameters"

($k \in \mathbb{R}$): If $\lambda \neq \frac{k}{2}(1 - \frac{k}{2})$ then $\pi(s_1, s_2, \varepsilon)$ will be irreducible. $P_k(\lambda, \varepsilon)$ Principal series.
 If $\lambda = \frac{k}{2}(1 - \frac{k}{2})$ then $\pi(s_1, s_2, \varepsilon)$ Not irreducible.

Description of $\pi(s_1, s_2, \varepsilon) = H(s_1, s_2, \varepsilon)$

$H(s_1, s_2, \varepsilon)$ is the Hilbert space completion of

$$H^\infty(s_1, s_2, \varepsilon) = \left\{ f \in C^\infty(G) \mid \begin{aligned} f\left(\begin{pmatrix} y & x \\ 1 & 1 \end{pmatrix} g\right) &= y^{s_1 + 1/2} y_2^{s_2 - 1/2} f(g), \quad y, x \in \mathbb{R}^+ \\ f\left(\begin{pmatrix} -1 & \\ & 1 \end{pmatrix} g\right) &= (-1)^\varepsilon f(g) \end{aligned} \right\}$$

• $G = GL(2, \mathbb{R})^+ \curvearrowright H(s_1, s_2, \varepsilon)$ by right translation:

$$(\pi(g)f)(x) = f(xg).$$

• $H^\infty(s_1, s_2, \varepsilon)$ Hermitian inner product:

$$\langle f_1, f_2 \rangle = \frac{1}{2\pi} \int_0^{2\pi} \int_0^\infty f_1(k\theta) \overline{f_2(k\theta)} dk d\theta$$

→ Remark: $f \in H^\infty(s_1, s_2, \varepsilon)$ is determined by its restriction to $k = S O(2)$
 → use Iwasawa decomp.

→ Moreover, as $f \in C^\infty(k)$ s.t. $f(k\theta + \pi) = (-1)^\varepsilon f(k\theta)$ can be extended to

$$\hat{f} \in H^\infty(s_1, s_2, \varepsilon) \text{ by defining } \hat{f}(g) = u^{s_1 + s_2} y^s f(k\theta)$$

$$\text{where } g = \begin{pmatrix} u & \\ & u \end{pmatrix} \begin{pmatrix} y & x \\ & y \end{pmatrix} k\theta, \quad k\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, \quad x, y, u \in \mathbb{R}^+, \theta \in \mathbb{R}.$$

Alternatively, $\pi(s_1, s_2, \varepsilon) = H(s_1, s_2, \varepsilon)$ is obtained by inducing ρ from $B(\mathbb{R})^+$ to $G = GL(2, \mathbb{R})^+$.

→ $\rho: B(\mathbb{R})^+ \rightarrow \mathbb{C}^\times$ is the 1D rep corresponding to the character χ .
 $\chi: B(\mathbb{R})^+ \rightarrow \mathbb{C}^\times, \quad \chi\left(\begin{pmatrix} y & x \\ & y \end{pmatrix}\right) = \text{sgn}(y)^\varepsilon |y|^{s_1} |y_2|^{s_2}$.

Def: Let G be a topological group, H closed subgroup. Let (ρ, V) be a rep of H .
 The space $V^G := \{ f: G \rightarrow V \mid f(hg) = \sqrt{\frac{\delta_H(h)}{\delta_G(g)}} \rho(h) f(g), \quad h \in H \}$.

• δ_H, δ_G : modular quasicharacters. $d_R(g) = \delta_G(g) d_L(g)$, Left-Right Invariant

Apply $G = GL(2, \mathbb{R})^+, \quad H = B(\mathbb{R})^+ = \left\{ \begin{pmatrix} y & x \\ & y \end{pmatrix} \in G = GL(2, \mathbb{R})^+ \right\}$ upper triangular

$$\delta_G = 1, \quad \delta_H\left(\begin{pmatrix} y & x \\ & y \end{pmatrix}\right) = y_1/y_2 \quad (\text{Exercise})$$

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Proposition 2.5.4: If $\ell = E(e)$, $\exists!$ $f_\ell \in H(s_1, s_2, E)$ s.t. $f_\ell(u_0) = e^{i\ell\theta}$
(then $f_\ell(g) = u^{s_1+s_2} y^s e^{i\ell\theta}$)

Then $Hf_\ell = \ell f_\ell$, $Rf_\ell = (s + \frac{1}{2})f_{\ell+2}$, $Lf_\ell = (s - \frac{1}{2})f_{\ell-2}$
and $\Delta f_\ell = \lambda f_\ell$, $Zf_\ell = \mu f_\ell$ where $\lambda = s(1-s)$, $\mu = s_1+s_2$, $s = \frac{1}{2}(s_1-s_2+1)$

Proof: Use $dZ = u \frac{\partial}{\partial u}$, $dR = e^{2i\theta} (iY \frac{\partial}{\partial x} + Y \frac{\partial}{\partial y} + \frac{1}{2i} \frac{\partial}{\partial \theta})$, $\partial H = -i \frac{\partial}{\partial \theta}$
 $dL = e^{-2i\theta} (-iY \frac{\partial}{\partial x} + Y \frac{\partial}{\partial y} - \frac{1}{2i} \frac{\partial}{\partial \theta})$

Compute: $-4\Delta f_\ell = (\ell^2 + 2(s + \frac{1}{2} - 1)(s - \frac{1}{2}) + 2(s - \frac{1}{2} - 1)(s + \frac{1}{2})) f_\ell$

Theorem 2.5.3: Let H be the $(\mathfrak{g}_0, K)$ -module of K -finite vectors in the Hilbert space $H(s_1, s_2, E)$. Let $s = \frac{1}{2}(s_1 - s_2 + 1)$, $\lambda = s(1-s)$, $\mu = s_1 + s_2$.
Then Δ & Z act as scalars on H , with eigenvalues λ & μ resp.

$P_M(E)$ (Discrete Series) (i) Suppose $s < \frac{1}{2}$ s.t. $s = \frac{k}{2}$ with $k \in \mathbb{Z}$, $k \equiv E(2)$. Then H is irreducible and the set of K -types of H consists of all integers congruent to E mod 2.

(ii) Suppose $s \geq \frac{1}{2}$ and that $s = \frac{k}{2}$, where $k \geq 1$ int, $k \equiv E(2)$. Then H is not irreducible but has two irreducible invariant subspaces H_+ and H_- .
The K -types of $H_\pm$ is $\Sigma^\pm(k)$.

$D_M^+(1)$ (Limit of discrete series) The quotient $H / (H_+ \oplus H_-)$ is irreducible, unless $k=1$, in which case 0.
Its set of K -types is $\Sigma^0(k)$.

(iii) Suppose $s \leq \frac{1}{2}$ and that $s = 1 - \frac{k}{2}$, where $k \geq 1$ int, $k \equiv E(2)$.
Then H has an invariant subspace H^0 whose set of K -types is $\Sigma^0(k)$.
Here H^0 is irreducible, unless $k=1$, in which case it is 0.

The quotient H/H^0 decomposes into two irreducible invariant subspaces H^+ and H^- ; the set of K -types of $H^\pm$ is $\Sigma^\pm(k)$.

Proof: H contains f_ℓ for all $\ell \in \mathbb{Z}$, $\ell \equiv E(2)$. So $H(k) \neq 0$ for all $k \in \mathbb{Z}$, $k \equiv E(2)$.
Obv, action of L, R , can identify invariant subspaces of H . Observe irred, Quotient has remaining K -types. $\square$

{From $G = GL(2, \mathbb{R})^*$ to $G = GL(2, \mathbb{R})$ }

Theorem 2.5.5 (Irreducible $(\mathfrak{g}_0, K)$ -modules for $GL(2, \mathbb{R})$): The following is a complete list of the irreducible $(\mathfrak{g}_0, K)$ -modules, where $\mathfrak{g} = \mathfrak{gl}(2, \mathbb{R})$ and $K = O(2)$.
Every irreducible admissible $(\mathfrak{g}_0, K)$ -module may be realized as the space of K -finite vectors in an admissible representation of $G = GL(2, \mathbb{R})^*$ on a Hilbert space.

(i) The finite-dimensional representations are obtained by tensoring the symmetric powers of the standard representation with the 1-D reps of the form $\chi \otimes \det$, where $\chi: \mathbb{R}^\times \rightarrow \mathbb{C}^\times$.

(ii) If $\chi_1, \chi_2: \mathbb{R}^\times \rightarrow \mathbb{C}^\times$ characters, s.t. $\chi_1 \chi_2^{-1}$ is not of the form $y \mapsto \text{sgn}(y)^k |y|^s$, $s=0$, $k \in \mathbb{Z}$, $k \equiv E(2)$, then we have the irred $(\mathfrak{gl}_2(\mathbb{R}), O(2))$ -module $\pi(\chi_1, \chi_2)$.

(iii) If $\mu \in \mathbb{R}$, $k \in \mathbb{Z}$, $k \geq 1$, then there are the representations $D_\mu(k)$, called discrete series if $k \geq 2$ and limit of discrete series if $k=1$. (Discrete series: $D_\mu(k) = D_\mu^+(k) \oplus D_\mu^-(k)$)